STPM Physics · Semester 1 · Chapter 11

THERMODYNAMICS

11.1 First Law of Thermodynamics

DEFINITIONS & LAWS

- **First law of thermodynamics**: the increase in the internal energy of a system is equal to the heat supplied to the system minus the work done by the system on its surroundings (conservation of energy).
- **Internal energy** of a system is the sum of the random kinetic energy and the potential energy of its molecules; for an ideal gas there is no intermolecular potential energy, so U depends only on the thermodynamic temperature.
- **Heat** is the energy transferred between a system and its surroundings because of a temperature difference. **Work** is the energy transferred when the gas expands or is compressed against an external pressure.
- **Molar heat capacity at constant volume** C_V is the heat required to raise the temperature of one mole of gas by 1 K at constant volume; **at constant pressure** C_p is the heat required for the same rise at constant pressure; C_p is greater than C_V because additional heat is needed for the work done in expansion.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
First law of thermodynamics	$\Delta U = Q - W$	ΔU = change in internal energy (J); Q = heat supplied to the gas (J), negative when heat leaves the gas; W = work done by the gas (J), negative when the gas is compressed
Change in internal energy	$\Delta U = \frac{f}{2} nR\Delta T$	for an ideal gas, whatever the process; f = number of degrees of freedom; n = number of moles; ΔT = temperature change (K)
Work done by gas	$W = \int p\, dV$ = area under the p–V graph; constant volume: $W = 0$; constant pressure: $W = p\Delta V = nR\Delta T$; isothermal: $W = nRT \ln \frac{V_2}{V_1}$; adiabatic: $W = \frac{p_1V_1 - p_2V_2}{\gamma - 1}$	W = work done by the gas (J), negative when the gas is compressed; the area under the p–V graph holds for every process; p = pressure (Pa); ΔV = volume change (m^3); n = number of moles; T = temperature (K); V_1, V_2 = initial and final volumes (m^3); p_1, p_2 = initial and final pressures (Pa); $\gamma = \frac{C_p}{C_V}$

Quantity / law	Formula	Symbols, units & conditions
Heat at constant volume	$Q = nC_V\Delta T$	C_V = molar heat capacity at constant volume (J mol^{-1} K^{-1}); n = number of moles; ΔT = temperature change (K)
Heat at constant pressure	$Q = nC_p\Delta T$	C_p = molar heat capacity at constant pressure (J mol^{-1} K^{-1}); n = number of moles; ΔT = temperature change (K)
Molar heat capacities	$C_V = \frac{f}{2}R$; $C_p = C_V + R$	Mayer's relation: $C_p = C_V + R$; R = molar gas constant (J mol^{-1} K^{-1}); f = number of degrees of freedom
Ratio of heat capacities	$\gamma = \frac{C_p}{C_V} = \frac{f+2}{f}$	γ = ratio of molar heat capacities (no unit); f = number of degrees of freedom

TIPS TO ANSWER

- State the first law with the meaning and sign of every symbol before substituting: Q positive for heat absorbed, W positive for work done by the gas.
- Decide the signs from the wording: compressed → W negative; heat released or lost → Q negative; temperature rises → ΔU positive.
- Calculation: $\Delta U = \frac{f}{2}nR\Delta T$ first, then find the missing quantity from $\Delta U = Q - W$; give the unit J and the sign (increase or decrease, absorbed or released).
- Explanations: link temperature change to internal energy (random kinetic energy of the molecules), and state that work is done by the gas only when its volume increases.

SECTION A TRAPS

Trap: Wrong sign convention: taking W as the work done on the gas while using $\Delta U = Q - W$. If the question defines W as the work done on the gas, then $\Delta U = Q + W$.

Trap: Assuming that heat supplied always raises the temperature; during an isothermal expansion the heat supplied is converted entirely into work and $\Delta U = 0$.

Trap: Using Q instead of ΔU to find the temperature change when the volume or pressure also changes; ΔU depends only on ΔT, and Q depends on the process.

TRICKS

Trick: ΔU depends only on the initial and final temperatures, so for any process between the same two temperatures ΔU is identical; only Q and W differ.

> **Trick:** A quick sign check: gas heated and expands, so Q is positive and W is positive; gas compressed rapidly with no heat flow, so W is negative and ΔU is positive (temperature rises).

11.2 Thermodynamic Processes

DEFINITIONS & LAWS

- **Isothermal process** is a process at constant temperature: ΔU = 0, so Q = W; heat absorbed is converted entirely into work done by the gas (pV = constant, Boyle's law).
- **Adiabatic process** is a process in which no heat enters or leaves the system: Q = 0, so ΔU = −W; an adiabatic expansion cools the gas and an adiabatic compression heats it.
- **Constant-pressure (isobaric) process**: W = pΔV, and ΔU = Q − pΔV. **Constant-volume (isochoric) process**: W = 0, so Q = ΔU, and all the heat supplied increases the internal energy.
- Work done by the gas equals the area under the p–V graph; in a cycle, the net work done equals the area enclosed by the loop (clockwise: net work done by the gas; anticlockwise: net work done on the gas).
- An adiabatic curve is steeper than an isothermal curve through the same point on a p–V graph because γ is greater than 1.
- **Reversible process** is one in which the system passes through a continuous series of equilibrium states and can be returned to the initial state with no change in the surroundings.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Isothermal process	$W = nRT \ln\frac{V_2}{V_1} = p_1V_1 \ln\frac{V_2}{V_1}$; pV = constant	W = work done by the gas (J); n = number of moles; T = constant temperature (K); V_1, V_2 = initial and final volumes (m^3); p_1 = initial pressure (Pa)
Adiabatic relations	pV^γ = constant; $TV^{\gamma-1}$ = constant; $T^\gamma p^{1-\gamma}$ = constant	$\gamma = \frac{C_p}{C_V}$ ($\frac{5}{3}$ monatomic, $\frac{7}{5}$ diatomic)
Adiabatic work	$W = \frac{p_1V_1 - p_2V_2}{\gamma - 1} = \frac{nR(T_1 - T_2)}{\gamma - 1} = -\Delta U$	W = work done by the gas (J); p_1, V_1, T_1 = initial state; p_2, V_2, T_2 = final state; n = number of moles; R = molar gas constant (J mol^{-1} K^{-1}); $\gamma = \frac{C_p}{C_V}$
Constant-pressure (isobaric) process	$W = p\Delta V = nR\Delta T$; $Q = nC_p\Delta T$	constant pressure; p = pressure (Pa); ΔV = volume change (m^3); ΔT = temperature change (K); C_p = molar heat capacity at constant pressure (J mol^{-1} K^{-1})
Constant-volume (isochoric) process	$W = 0$; $Q = \Delta U = nC_V\Delta T$	constant volume; C_V = molar heat capacity at constant volume (J mol^{-1} K^{-1}); ΔT = temperature change (K)

Quantity / law	Formula	Symbols, units & conditions
Mayer's relation	$C_p - C_V = R$	C_p, C_V = molar heat capacities at constant pressure and volume (J mol^{-1} K^{-1}); R = molar gas constant (J mol^{-1} K^{-1})
First law for each process	$Q = \Delta U + W$	Q = heat supplied to the gas (J); ΔU = change in internal energy (J); W = work done by the gas (J)

TIPS TO ANSWER

- For each process, state the defining condition (ΔT = 0, Q = 0, Δp = 0 or ΔV = 0), then write the reduced form of the first law before substituting.
- Calculation: use pV = nRT to find the missing p, V or T, then W from the area or formula, $\Delta U = \frac{f}{2} nR\Delta T$, and Q = ΔU + W; show every step with units.
- p–V sketch: label axes, mark the initial and final states with arrows, draw a horizontal line (constant pressure), a vertical line (constant volume), a hyperbola (isothermal) and a steeper curve (adiabatic).
- For a cycle, tabulate Q, W and ΔU for each stage; ΔU over a complete cycle is zero, so the net heat supplied equals the net work done.

SECTION A TRAPS

Trap: Confusing isothermal and adiabatic: isothermal means ΔU = 0 with Q and W non-zero; adiabatic means Q = 0 with ΔU and W non-zero.

Trap: Applying pV^{γ} = constant to an isothermal process, or $W = nRT \ln\frac{V_2}{V_1}$ to an adiabatic process; each formula belongs to one process only.

Trap: Taking W = pΔV for a process in which the pressure changes; this holds only for constant-pressure changes, otherwise use the area under the graph.

TRICKS

Trick: Reading a p–V graph: the horizontal line is at constant pressure, the vertical line is at constant volume, and of two curves from the same point the steeper one is adiabatic.

Trick: At the same volume change from the same start, an adiabatic expansion ends at a lower pressure and a lower temperature than an isothermal expansion.